JONATHAN BHERI

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TECHNICAL SKILLS

Programming Languages: Structured text |Ladder Logic| Python | C++| MATLAB | SQL |Go lang
Industrial platforms: Allen-Bradley and Siemens PLC | Fanuc Robots | MES | Machine Vision | Servos and VFDs
Design and analysis: AutoCAD | ANSYS | CATIA | Unigraphics NX | Solidworks
Development platforms: Linux, Foreman , Docker, Jenkins, Electrical testing (ICT and FCT)
Machine Learning libraries: Scikit-learn | TensorFlow | Keras | OpenCV 3 |
Parallel Programming (GPU): CUDA **Web**: HTML | CSS | REST API | Flask | Nginx

EDUCATION

M.S

Master of Science in **Electrical and Computer Engineering** at **Colorado State University** **May '18**
Concentration in ***Robot control and vision***

Coursework: Machine Learning and adaptive Filtering, Computer vision, Non-linear control systems, Digital signal Processing, Digital image processing, Robot control and simulation, Advanced robotics: Redundancy and Optimization, Bio-inspired robots and Embedded systems

B.Tech

Bachelor of Technology in **Mechatronics** at **Mahatma Gandhi Institute of Technology, India** **May '15**
Concentration in ***Mechanical***

Coursework: IC applications, Microprocessors and microcontrollers, Control systems, Mechanics of Solids, Production Technology, CAD/CAM (Auto CAD, ANSYS), Dynamics of Robotics and C/C++ programming

EXPERIENCE

UnitX

Principal Solutions Architect, UnitX, Santa Clara, California **June '25-Oct '25**
• Managed two teams (Software and Hardware) as the technical sales team supporting pre and post sales till commissioning of AI Vision Systems.
• Architected and designed solutions for lensing, fixturing, and imaging for object detection for complex products and lines
• Helped develop programs and software scripts for custom customer solutions for defect detection to achieve less than 3% False positive rates.
• Key role in standardizing solutions across the various product solutions.
• Visited 17 different customer sites to understand needs and provide solutions.

Sigray

Principal Controls Engineer, Sigray, Concord, California **Jan '25-Oct '25**
• Owner for all controls activities across the company, encompassing integration and infrastructure for all the X-ray product lines
• Architected and implemented infrastructure design for system management and provisioning using Red Hat Enterprise Linux company-wide using Satellite
• Designed a high precision timing system with Piezo stages (PI USA) and timing/signal controller (Zebra-Quantum detector) for X-ray tomography imaging
• Engineered a highly sensitive leak detection system with comprehensive traceability for the XRM product line, enhancing product reliability

Enovix

Principal Equipment and Controls Engineer, Enovix , Fremont, California **Aug '22-Nov'24**
• Equipment and Controls Ownership: As the owner of the Degas, Formation, and Testing lines in Li-Si 3D Battery Fab, managed critical equipment and control systems. This included overseeing their installation, configuration, and commissioning.
• Team Leadership and Production Line Development: Successfully led multiple teams in constructing a state-of-the-art production line for battery formation and testing.
• Orchestrated project timelines, ensuring seamless execution during the construction of the production line—from its inception to achieving a throughput of 250 units per hour (UPH).
• Cutting-Edge Fire Safety Systems: Designed and implemented advanced fire safety systems specifically tailored for battery-related hazards. These systems prevent, detect, and effectively control battery fires, ensuring the safety of personnel and equipment. A 7-Fold Fire detection and control system was deployed.

EXPERIENCE

Enovix

• Automated Storage and Retrieval System (ASRS): Pioneered the design of an ASRS system for efficient battery management within the formation and testing warehouse. This system optimizes storage, retrieval, and flow, enhancing overall productivity. This design increase throughput by 12 % and 37% labor savings
• Data Pipelines and Quality Standards: Established robust data pipelines to collect information from diverse charging and testing equipment. This data-driven approach ensures adherence to performance and quality standards. Various ETLs, API and other accommodations were made for data management.
• Test Plan Execution: Developed comprehensive test plans and executed them meticulously. Accurate data collection and analysis were central to validating equipment performance and reliability.
• Collaboration and Goal Achievement: Collaborated closely with cross-functional teams to achieve ambitious production targets of 250 UPH. Safety and reliability remained our guiding principles throughout the process

Tesla

Manufacturing Equipment Engineer, Tesla Battery pack Gigafactory, Sparks,Nevada Nov ’18 – July ‘22
• Lead several projects for improving the OEE, Quality, Safety and Cycle time of machine on Battery pack and Enclosure lines using Allen Bradley PLCs and Ignition SCADA(HMI).
• Helped in decreasing the Tact time by 15 sec. on pack line which amounts to an additional throughput of 147,000 packs per annum or 37% from the factory.
• Developed and updated logic on numerous machines for product development and carried out many continuous improvement projects on both the lines. And made complimentary HMI UI and scripts on Ignition.
• Battery pack and BMS system End of line (EOL) Tester sustaining and improvements. Designed connectors with better modularity and repairability which improved the design for mass manufacturing.
• Develop scripts and setup Linux services for Electrical testing on battery pack and BMS
• Coordinated and developed processes for multiple Equipment and New Product Introduction (NPI)
• Used Kubernetes containerized scripts to automate several production critical tasks
• Deployed safety systems for the production line using light curtains, area scanners, E-stops and pull cords.
• Worked with roller conveyor systems using Allen Bradley Powerflex 523,525 and 527 controllers
• Designed a vacuum suction system for robust and failure redundant transfer of material by a robot.
• Deployed several fastening and fastener positioning systems using Atlas Copco’s Power focus, Power mac, TPS and ILG systems.
• Developed a system for process control of Airway and Böllhoff feeders’ inventory across the factory
• Worked on various vision and scanners projects for process control using Keyence XGX, IV, CVX, SR and Cognex DM systems. Helped in setting up a novel Raspberry Pi vision system for FOD detection.
• Collaborated with cross functional team in conducting several DOEs, feasibility analysis and data analytics for understanding and optimizing the line parameters using Python, Minitab and Tableau and gathering data from various sources such as SQL databases, REST API and on the line data.
• Worked with various electrical and leak testing equipment for checking Battery pack integrity and health.
• Setup several non-conforming material and repair stations and standardized the data collection methods.
• Designed and prototyped custom equipment and tools using CAD and various machining techniques
• Helped in development of factory wide training and Manufacturing Instruction deployment system
• Trained several engineers and technicians on the line required systems and created training documentation

Vizag Steel Plant

Automation Intern at **Rashtriya Ispat Nigam Limited**, **Vizag Steel Plant** **May ’14 – June ’14**
• Performed inspection of the fuel and raw material delivery system regarding functioning of the blast furnace
• Assessed the system model and developed a system to improve the performance of the blast furnace.
• Developed the Failure Tolerant control system to control the performance of the blast furnace using the flow of fuel and raw materials into the system.
• Reported the assessment on control system to the Instrumentation Engineer for implementing the system model.